Exact classification of rank-one orthogonal projections that are ℓ^∞-contractive

Candidate full-solution packet for arXiv:0903.3580

11 August 2026

Status: candidate full solution, likely valid, subject to human review. For a unit vector $v \in \mathbb{C}^N$, the orthogonal projection $P_v = vv^*$ is ℓ^∞-contractive if and only if all nonzero coordinates of v have the same modulus. Applying this dimension-free criterion to the spherical coordinates in arXiv:0903.3580 gives an exact parameter list and proves that no pairs beyond the families already observed in the source exist.

1 Source question

Mugnolo studies heat semigroups on a three-edge star with one-dimensional boundary space $Y_{\xi,\phi} \subset \mathbb{C}^3$. In the real spherical chart, this space is generated by

$$v_{\xi,\phi} = (\sin\xi\cos\phi,\ \sin\xi\sin\phi,\ \cos\xi), \qquad \xi,\phi \in [0,\pi), \tag{1}$$

and its orthogonal projection is $P_{Y_{\xi,\phi}} = v_{\xi,\phi}v_{\xi,\phi}^{\mathsf{T}}$. The source reduces L^∞-contractivity of the associated heat semigroup (with $S = 0$) to ℓ^∞-contractivity of this matrix, plots the three absolute row sums, and verifies several isolated points and decoupled families. It then states [1, p. 19]:

> "We do not know whether further pairs (ξ, ϕ) leading to L^∞-contractivity exist."

The relevant source pages are reproduced in Appendix A.

Idea of the proof

There is no need to solve three trigonometric inequalities. The absolute row sum of the rank-one matrix vv^* is $|v_i|\sum_j |v_j|$, so its largest row sum is $\|v\|_\infty\|v\|_1$. If v is a unit vector and $m = \|v\|_\infty$, then

$$1 = \sum_i |v_i|^2 \le m \sum_i |v_i|.$$

Contractivity forces equality. Its defect is the sum of the nonnegative terms $|v_i|(m - |v_i|)$, so equality occurs exactly when each coordinate has modulus either zero or m. The spherical parameter list then follows by separating supports of sizes one, two, and three.

2 A dimension-free classification

Theorem 1. *Let $v = (v_1, \dots, v_N) \in \mathbb{C}^N$ satisfy $\|v\|_2 = 1$, and let $P_v x = \langle x, v\rangle v$ be the orthogonal projection onto $\mathbb{C}v$. Then*

$$\|P_v\|_{\ell^\infty_N \to \ell^\infty_N} = \|v\|_\infty \|v\|_1 \ge 1. \tag{2}$$

Moreover, the following are equivalent:

1. *P_v is ℓ^∞-contractive;*
2. *every nonzero coordinate of v has the same modulus;*
3. *for some nonempty support $S \subset \{1, \dots, N\}$,*

$$|v_i| = |S|^{-1/2} \quad (i \in S), \qquad v_i = 0 \quad (i \notin S).$$

Proof. The (i, j) entry of $P_v = vv^*$ has modulus $|v_i||v_j|$. The induced ℓ^∞ norm is the largest absolute row sum, hence

$$\|P_v\|_{\infty\to\infty} = \max_i \sum_j |v_i||v_j| = \|v\|_\infty \|v\|_1.$$

Put $m = \|v\|_\infty$. Since $|v_i| \le m$,

$$1 = \sum_i |v_i|^2 \le m \sum_i |v_i| = \|P_v\|_{\infty\to\infty}. \tag{3}$$

Thus contractivity is equivalent to equality in (3). But the defect is

$$m \sum_i |v_i| - \sum_i |v_i|^2 = \sum_i |v_i|(m - |v_i|),$$

a sum of nonnegative numbers. It vanishes exactly when $|v_i| \in \{0, m\}$ for every i. If the nonzero support has size k, unit normalization gives $km^2 = 1$, hence $m = k^{-1/2}$. The converse is immediate. □

The theorem also permits arbitrary complex phases on the nonzero coordinates. For the source's real spherical family, only signs occur.

3 Complete answer in the source coordinates

Let

$$\alpha := \arctan \sqrt{2} \in (0, \pi/2).$$

Corollary 1. *For $(\xi, \phi) \in [0, \pi)^2$, the projection $P_{Y_{\xi,\phi}}$ in the source is ℓ^∞-contractive if and only if (ξ, ϕ) belongs to*

$$\big(\{0\} \times [0, \pi)\big) \tag{4}$$
$$\cup \left(\left\{ \frac{\pi}{2} \right\} \times \left\{ 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4} \right\} \right) \tag{5}$$
$$\cup \left(\left\{ \frac{\pi}{4}, \frac{3\pi}{4} \right\} \times \left\{ 0, \frac{\pi}{2} \right\} \right) \tag{6}$$
$$\cup \left(\{\alpha, \pi - \alpha\} \times \left\{ \frac{\pi}{4}, \frac{3\pi}{4} \right\} \right). \tag{7}$$

Consequently there are no further pairs in the chart considered by the source. With $S = 0$, this is also the exact list of the corresponding L^∞-contractive heat semigroups, for both the time-independent and dynamic boundary problems covered by the source's equivalence theorem.

Proof. Apply Theorem 1 to (1) and separate by support size.

If exactly one coordinate is nonzero, one obtains $\xi = 0$ with arbitrary ϕ, or

$$(\xi, \phi) = (\pi/2, 0) \quad \text{or} \quad (\pi/2, \pi/2).$$

If exactly two coordinates are nonzero and have equal absolute value, the support $\{1, 2\}$ gives

$$\xi = \frac{\pi}{2}, \qquad \phi \in \left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\};$$

the support $\{1, 3\}$ gives

$$\phi = 0, \qquad \xi \in \left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\};$$

and the support $\{2, 3\}$ gives

$$\phi = \frac{\pi}{2}, \qquad \xi \in \left\{\frac{\pi}{4}, \frac{3\pi}{4}\right\}.$$

If all three coordinates are nonzero with equal absolute value, then $|\cos\phi| = |\sin\phi|$, so $\phi \in \{\pi/4, 3\pi/4\}$. Equality with the third coordinate then becomes

$$|\sin\xi|/\sqrt{2} = |\cos\xi|,$$

or $|\tan\xi| = \sqrt{2}$. Hence $\xi \in \{\alpha, \pi - \alpha\}$. Collecting the cases gives exactly (4). $\square$

Modulo the coordinate degeneracy at $\xi = 0$, list (4) represents thirteen real lines: three coordinate axes, six equal-modulus two-coordinate lines, and four equal-modulus three-coordinate lines. The source had already exhibited these families; Corollary 1 supplies the missing exhaustiveness proof.

4 Verification, scope, and novelty

The argument is algebraic and exact. The companion script `code/verify_classification.py` checks the norm formula on random vectors, verifies all displayed parameter representatives, and confirms that small perturbations of every nondegenerate solution have norm greater than one. These computations are sanity checks, not part of the proof.

This fully answers the printed question for one-dimensional boundary spaces of $\mathbb{C}^3$ and strengthens it to rank-one orthogonal projections in every finite dimension. It does not classify higher-rank orthogonal projections, nor does it settle the distinct question in the source about the complements $I - P_{Y_{\xi,\phi}}$.

The bounded novelty check on 11 August 2026 searched the run registry and its solution, attempt, and proof-gap indexes for the arXiv id, title, and central projection/contractivity phrases. Exact-title, exact-question, and rank-one-projection primary-source web searches found the source paper and general literature on contractive projections, but no later paper giving this parameter classification or answering the quoted question. Because the argument is elementary, this is not an exhaustive priority claim.

Human-review recommendation. Check that the source's reduction from semigroup contractivity to ℓ^∞-contractivity of $P_{Y_{\xi,\phi}}$ is being used with $S = 0$, as in its example section. The finite-dimensional classification itself is independent of the PDE setup.

A Source-page evidence

Now, consider a semi-infinite star with only two edges, i.e., $H = \mathbb{C}^2$. Neglecting the trivial (uncoupled) boundary conditions defined by $Y = \{0\}$ and $Y = \mathbb{C}^2$ we can consider all 1-dimensional subspaces $\mathcal{Y} \equiv Y_\xi$ of $\mathbb{C}^2$ by means of the parametrisation

$$P_{Y_\xi} := \begin{pmatrix} \cos^2\xi & \sin\xi\ \cos\xi \\ \sin\xi\ \cos\xi & \sin^2\xi \end{pmatrix}, \qquad \xi \in [0,\pi),$$

where Y_ξ denotes the range of the orthogonal projection P_{Y_ξ}. Observe that $\xi = 0$, $\xi = \frac{\pi}{4}$, $\xi = \frac{\pi}{2}$ and $\xi = \frac{3\pi}{4}$ correspond to uncoupled Dirichlet/Neumann, to Kirchhoff, to uncoupled Neumann/Dirichlet and to anti-Kirchhoff boundary conditions, respectively, as can be checked directly.

We are going to discuss the submarkovian property of the semigroup associated with these subspaces in dependence of ξ. A direct computation shows that the semigroup $(e^{t\Delta_{Y_\xi,0}})_{t\geq 0}$ is positive if and only if $\xi \in [0, \frac{\pi}{2}]$. Furthermore, by Theorem 3.6 the semigroup that governs (NDBC) is $L^\infty(\Omega\times\Xi;\mathbb{C})$-contractive if and only if P_{Y_ξ} is $L^\infty(\Xi;\mathbb{C})$-contractive , i.e., if and only if the inequalities

$$\cos^2\xi + |\sin\xi\ \cos\xi| \leq 1 \qquad \text{and} \qquad |\sin\xi\ \cos\xi| + \sin^2\xi \leq 1$$

hold simultaneously. The former (resp., the latter) inequality holds if and only if $\xi \notin (0,\frac{\pi}{4}) \cup (\frac{3\pi}{4},\pi)$ (resp., if and only if $\xi \notin (\frac{\pi}{4},\frac{\pi}{2}),(\frac{\pi}{2},\frac{3\pi}{4})$).

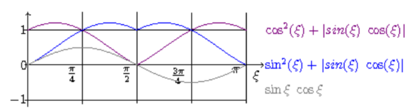

FIGURE 1

Therefore, the L^∞-contractivity of the semigroup associated with Kirchhoff boundary conditions represents a singularity. In particular, a submarkovian semigroup is generated *exactly* in the following five cases:

- with uncoupled Dirichlet/Dirichlet boundary conditions,
- with uncoupled Neumann/Neumann boundary conditions,
- with uncoupled Dirichlet/Neumann boundary conditions,
- with uncoupled Neumann/Dirichlet boundary conditions and finally
- with Kirchhoff boundary conditions.

Similarly, we can consider general boundary conditions defined by 1-dimensional subspaces of H for a semi-infinite star with 3 edges ($H = \mathbb{C}^3$). They can be investigated by means of spherical boundary conditions, i.e., considering spaces $\mathcal{Y} \equiv Y_{\xi,\phi}$ that are ranges of the orthogonal projections

$$P_Y \equiv P_{Y_{\xi,\phi}} = \begin{pmatrix} \sin^2\xi\ \cos^2\phi & \sin^2\xi\ \sin\phi\ \cos\phi & \sin\xi\ \cos\xi\ \cos\phi \\ \sin^2\xi\ \sin\phi\ \cos\phi & \sin^2\xi\ \sin^2\phi & \sin\xi\ \cos\xi\ \sin\phi \\ \sin\xi\ \cos\xi\ \cos\phi & \sin\xi\ \cos\xi\ \sin\phi & \cos^2\xi \end{pmatrix}, \qquad \xi,\phi \in [0,2\pi). \tag{4.1}$$

Analysing the behaviour of $P_{Y_{\xi,\phi}}$ in dependence of ξ,ϕ as done above for P_{Y_ξ} is less elementary. While the matrix is clearly positive if and only if $\xi,\phi \in [0,\frac{\pi}{2}] \cup [\pi,\frac{3\pi}{2}]$, it is not clear how to determine all the values ξ,ϕ leading to L^∞-contractivity, i.e., all the values ξ,ϕ such that the three functions

$$\begin{aligned} &\sin^2\xi\ \cos^2\phi + |\sin^2\xi\ \sin\phi\ \cos\phi| + |\sin\xi\ \cos\xi\ \cos\phi|, \\ &|\sin^2\xi\ \sin\phi\ \cos\phi| + \sin^2\xi\ \sin^2\phi + |\sin\xi\ \cos\xi\ \sin\phi| \quad \text{and} \qquad\qquad \xi,\phi\in[0,\pi) \\ &|\sin\xi\ \cos\xi\ \cos\phi| + |\sin\xi\ \cos\xi\ \sin\phi| + \cos^2\xi \end{aligned}$$

are simultaneously ≤ 1, corresponding to the three conditions for L^∞-contractivity associated with the three rows of the matrix $P_{Y_{\xi,\phi}}$ in (4.1).

Figure 1: Printed page 18 of arXiv:0903.3580, introducing the spherical projection and reducing the question to three absolute row-sum inequalities.

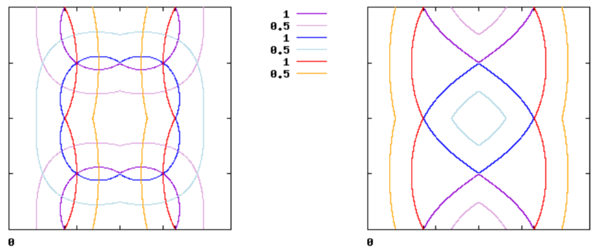

FIGURE 2A FIGURE 2B

In Figure 2A we have plotted[7] the level lines of the above functions for the value 1 (in violet, blue and red, respectively). This suggests that the ten parameter choices

- $(\xi, \phi) = (\arctan \sqrt{2}, \frac{\pi}{4})$,
- $(\xi, \phi) = (\pi - \arctan \sqrt{2}, \frac{\pi}{4})$,
- $(\xi, \phi) = (\arctan \sqrt{2}, \frac{3\pi}{4})$,
- $(\xi, \phi) = (\pi - \arctan \sqrt{2}, \frac{3\pi}{4})$.
- $(\xi, \phi) = (\frac{\pi}{4}, \frac{\pi}{2})$,
- $(\xi, \phi) = (\frac{3\pi}{4}, \frac{\pi}{2})$,
- $(\xi, \phi) = (\frac{\pi}{2}, \frac{\pi}{4})$,
- $(\xi, \phi) = (\frac{\pi}{2}, \frac{3\pi}{4})$,
- $(\xi, \phi) = (\frac{\pi}{4}, 0)$,
- $(\xi, \phi) = (\frac{3\pi}{4}, 0)$,

lead to an L^∞-contractive semigroup – as in fact can be checked directly.

Observe that $Y_{\xi,\phi}$ identifies Kirchhoff boundary conditions if and only if

$$P_{Y_{\xi,\phi}} = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

i.e., if and only if $\xi = \arctan \sqrt{2}$ and $\phi = \frac{\pi}{4}$. A direct computation shows that the remaining nine cases correspond to boundary conditions defined by means of spaces Y given by

$$\{(c,-c,-c) : c \in \mathbb{C}\}, \quad \{(c,c,-c) : c \in \mathbb{C}\}, \quad \{(c,-c,c) : c \in \mathbb{C}\},$$
$$\{(c,c,0) : c \in \mathbb{C}\}, \quad \{(c,-c,0) : c \in \mathbb{C}\}, \quad \{(0,c,c) : c \in \mathbb{C}\},$$
$$\{(0,c,-c) : c \in \mathbb{C}\}, \quad \{(c,0,c) : c \in \mathbb{C}\}, \quad \{(c,0,-c) : c \in \mathbb{C}\}.$$

While the last six subspaces only describe some decoupling of any of the three edges, we cannot find any physical interpretation for the first three boundary conditions. One can see that analogous boundary conditions give rise to L^∞-contractive semigroups also in higher dimensional spaces $H = \mathbb{C}^N$ for any $N \in \mathbb{N}$.

It ought to be remarked that not all relevant values become evident through the above plot: one can see that decoupled boundary conditions arise with $Y_{0,\phi}$ and $Y_{\frac{\pi}{2},\phi}$ for all $\phi \in [0,\pi)$ as well as with $Y_{\xi,0}$ and $Y_{\xi,\frac{\pi}{2}}$ for all $\xi \in [0,\pi)$. Hence, using again the computations performed in the case of $H = \mathbb{C}^2$, we see that $Y_{\frac{\pi}{2},\phi}$ lead to L^∞-contractivity for $\phi \in \{0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}\}$, and so do $Y_{\xi,0}$ and $Y_{\xi,\frac{\pi}{2}}$ for $\xi \in \{0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}\}$ as well as $Y_{0,\phi}$ for all $\phi \in [0,\pi)$. We do not know whether further pairs (ξ,ϕ) leading to L^∞-contractivity exist.

[7] The figure has been obtained using Gnuplot 4.2 with a grid density of 1000 on both axes. For reference we have plotted the level lines for the value 0.5, too. On the ξ-axis (horizontal) we have highlighted the values $\frac{\pi}{4}$, $\arctan\sqrt{2}$, $\frac{3\pi}{4}$ and $\pi - \arctan\sqrt{2}$. On the ϕ-axis (vertical) we have highlighted the values $\frac{\pi}{4}$, $\frac{\pi}{2}$ and $\frac{3\pi}{4}$.

Figure 2: Printed page 19 of arXiv:0903.3580, listing the observed solutions and asking whether further L^∞-contractive parameter pairs exist.

References

[1] Delio Mugnolo, *Vector-valued heat equations and networks with coupled dynamic boundary conditions*, arXiv:0903.3580, 2009.